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Software corp

Smart Parking Platform

Existing Software Landscape

Real time parking platforms:

1. Park Houston - allows for reservation of street parking
2. Park Mobile - allows payment of street parking in certain zones
3. Downtown Houston Website - Allows for parking reservations in lots
4. Ideal Parking Management - Management service that provides lot operators with tools to control their parking lot

How our application will be different

The applications listed above are great, but do not offer the solution that many are looking for. Our application will have the ability to estimate parking spot capacity for street parking, as that can be variable, but will display exact lot/ garage capacity, for easier parking. The others also do not include integrated directions for guidance to spots, giving us a competitive edge that makes our application more modern comparatively.

SRS Draft

- Software Requirements Specification (SRS)
- Project: Smart Parking Platform
- Version: 2.0
- Date: 09.10.2026

• 1. Introduction

This Smart Parking Platform (S.P.P.) will be brought to you by \$oftware ¢orp. We are a development studio aimed at delivering solutions to real world problems, by making lives easier one app at a time. The project to create a S.P.P. was acquired through a close partnership with the City of Houston to create a solution to the parking frustrations Houstonians face.

• 1.1 Purpose

- This document defines the requirements for the Smart Parking Platform

• 1.2 Scope

- In scope is to create an interactive application that will allow users to view the status of parking spaces in real time. For operators of parking spots, this will streamline parking spot payment and tracking.
- Out of scope will be any type of physical hardware such as parking gates and cameras and general street parking.

• 2. Overall Description

- Users: Drivers and Vehicle Owners, Parking Lot Operators, Payment Services, Finance and Billing Services, City parking departments, System Admins
- The application will have the ability to be run on Mobile, and the Web.
- The mobile devices will need to have access to the App Store for ios devices, and the Google Play Store for Android devices.
- Assumptions: Internet Access, modern mobile device or computer, current software version of user device for app compatibility

3. Functional Requirements

- FR1: The system shall allow users to register and log in securely using email or Google/Apple authentication.
- FR2: The system shall show current availability of parking spaces within an interactive map.
- FR3: The system shall allow users to reserve parking spaces for their desired amount of time.
- FR4: The system shall handle digital payments, and give the user a receipt for proof of purchase.
- FR5: The system shall send notifications to let the user know their parking reservation will expire soon.
- FR6: The system shall give parking operators a view of current capacity and money generated in their desired timeframe.
- FR7: The system shall allow operators to adjust prices of parking spots at their discretion.
- FR8: The system shall link with third party navigation systems, such as Google Maps, to give directions to the space.

• 4. Non-Functional Requirements

- NFR1: The system shall support at least 1,000 concurrent users.
- NFR2: The system shall update parking spot availability within 1 second of a change in occupancy.
- NFR3: The system shall provide 99.99% uptime.
- NFR4: The system shall process payments within 5 seconds.

• 5. Use Case Examples

5.1

Title: Register a new user account

Actor: Driver

Precondition: User has downloaded the app and has a valid email address

Steps:

1. User selects "Sign Up"
2. User enters email, password, and vehicle information
3. System sends a verification email
4. User confirms email and account is activated

Postcondition: User can log in and access the platform

5.2

Title: Reserve a parking space

Actor: Driver

Precondition: User is logged in and has a valid payment method on file

Steps:

1. User opens the map and selects an available parking space
2. User selects a reservation time window
3. User confirms payment
4. System reserves the space and sends a confirmation with directions

Postcondition: Space is marked unavailable to other users for the reserved window

5.3

Title: Cancel a reservation

Actor: Driver

Precondition: User has an active reservation

Steps:

1. User navigates to "My Reservations"
2. User selects the reservation to cancel
3. User confirms cancellation
4. System processes any applicable refund per cancellation policy

Postcondition: Space is released and marked available for other users

5.4

Title: View real time parking availability

Actor: Driver

Precondition: User has location services enabled and internet access

Steps:

1. User opens the app
2. System retrieves current location
3. System displays nearby parking facilities on an interactive map
4. Map updates availability in real time

Postcondition: User can view up to date open spaces near them

5.5

Title: Get navigation directions to a reserved space

Actor: Driver

Precondition: User has an active reservation

Steps:

1. User selects "Navigate" from reservation details
2. System pulls route data from the mapping service
3. System displays turn by turn directions
4. User follows route to the facility

Postcondition: User arrives at the correct facility and space

5.6

Title: Process a digital payment

Actor: Driver

Precondition: User has confirmed a reservation and has a saved payment method

Steps:

1. System calculates total cost based on time and facility rate
2. User confirms the charge
3. System sends transaction to third party payment provider
4. Payment provider returns confirmation

Postcondition: Transaction is completed and a digital receipt is issued

5.7

Title: View reservation history and receipts

Actor: Driver

Precondition: User is logged in and has at least one past reservation

Steps:

1. User navigates to "History"
2. System displays list of past reservations
3. User selects a specific reservation
4. System displays receipt details

Postcondition: User can view or export a past transaction record

5.8

Title: Receive a reservation expiration notification

Actor: Driver

Precondition: User has an active reservation nearing its end time

Steps:

1. System monitors time remaining on active reservations
2. System detects reservation is within 15 minutes of expiring
3. System sends a push notification to the user
4. User can choose to extend or end the reservation

Postcondition: User is alerted before losing the reserved space

5.9

Title: Extend an active reservation

Actor: Driver

Precondition: User has an active reservation and the space has no conflicting reservation

Steps:

1. User selects "Extend Reservation"
2. System checks space availability for the extended window
3. User confirms additional payment
4. System updates the reservation end time

Postcondition: Reservation is extended and space remains held

5.10

Title: Register a parking facility (operator onboarding)

Actor: Parking Facility Operator

Precondition: Operator has a valid business account request

Steps:

1. Operator submits facility details and documentation
2. System routes request to administrative approval queue
3. Administrator reviews and approves the facility
4. System activates the facility and its spaces in the platform

Postcondition: Facility appears in the app for drivers to reserve

5.11

Title: Adjust pricing based on demand

Actor: Parking Facility Operator

Precondition: Operator is logged into the administrative dashboard

Steps:

1. Operator navigates to "Pricing Settings"
2. Operator selects a time block or demand condition
3. Operator sets a new rate
4. System applies the updated rate to future reservations

Postcondition: New pricing is reflected for drivers making reservations

5.12

Title: Monitor real time occupancy

Actor: Parking Facility Operator

Precondition: Operator is logged into the administrative dashboard

Steps:

1. Operator opens the "Occupancy" view
2. System displays current occupied vs. available spaces
3. Operator filters by zone or level
4. Dashboard updates counts in real time

Postcondition: Operator has an accurate live view of facility usage

5.13

Title: Generate an occupancy and revenue report

Actor: Parking Facility Operator

Precondition: Operator is logged into the administrative dashboard and facility has reservation history

Steps:

1. Operator selects "Reports"
2. Operator chooses a date range and report type
3. System compiles occupancy and revenue data
4. System generates a downloadable report

Postcondition: Operator receives a report for business analysis

5.14

Title: Resolve a billing problem

Actor: Billing/Finance Personnel

Precondition: A user has submitted a dispute for a charge

Steps:

1. Billing staff reviews the dispute ticket
2. Staff cross references transaction and reservation logs

3. Staff determines validity of the dispute
 4. Staff issues a refund or denies the claim with explanation
- Postcondition: Dispute is resolved and user is notified of the outcome

5.15

Title: Suspend a facility for policy violation

Actor: System Administrator

Precondition: Administrator has identified a facility violating platform policy

Steps:

1. Administrator opens the facility's account in the admin panel
 2. Administrator reviews violation details
 3. Administrator selects "Suspend Facility"
 4. System disables the facility's listings and notifies the operator
- Postcondition: Facility no longer appears to drivers until issue is resolved

Workbreaddown Structure

1. Requirements & Planning

- 1.1 Competitive & Market Research
- 1.2 Vision & Scope Definition
- 1.3 Software Requirements Specification
- 1.4 Requirements Validation & Stakeholder Approval
 - 1.4.1 Review requirements
 - 1.4.2 Present SRS draft to stakeholders
 - 1.4.3 Obtain formal approval

2. Design

- 2.1 System Architecture Design
- 2.2 UI/UX Design
- 2.3 Database & API Design
- 2.4 Design Review & Validation
 - 2.4.1 Conduct architecture walkthrough
 - 2.4.2 Conduct UI/UX design review
 - 2.4.3 Validate API contracts against requirements

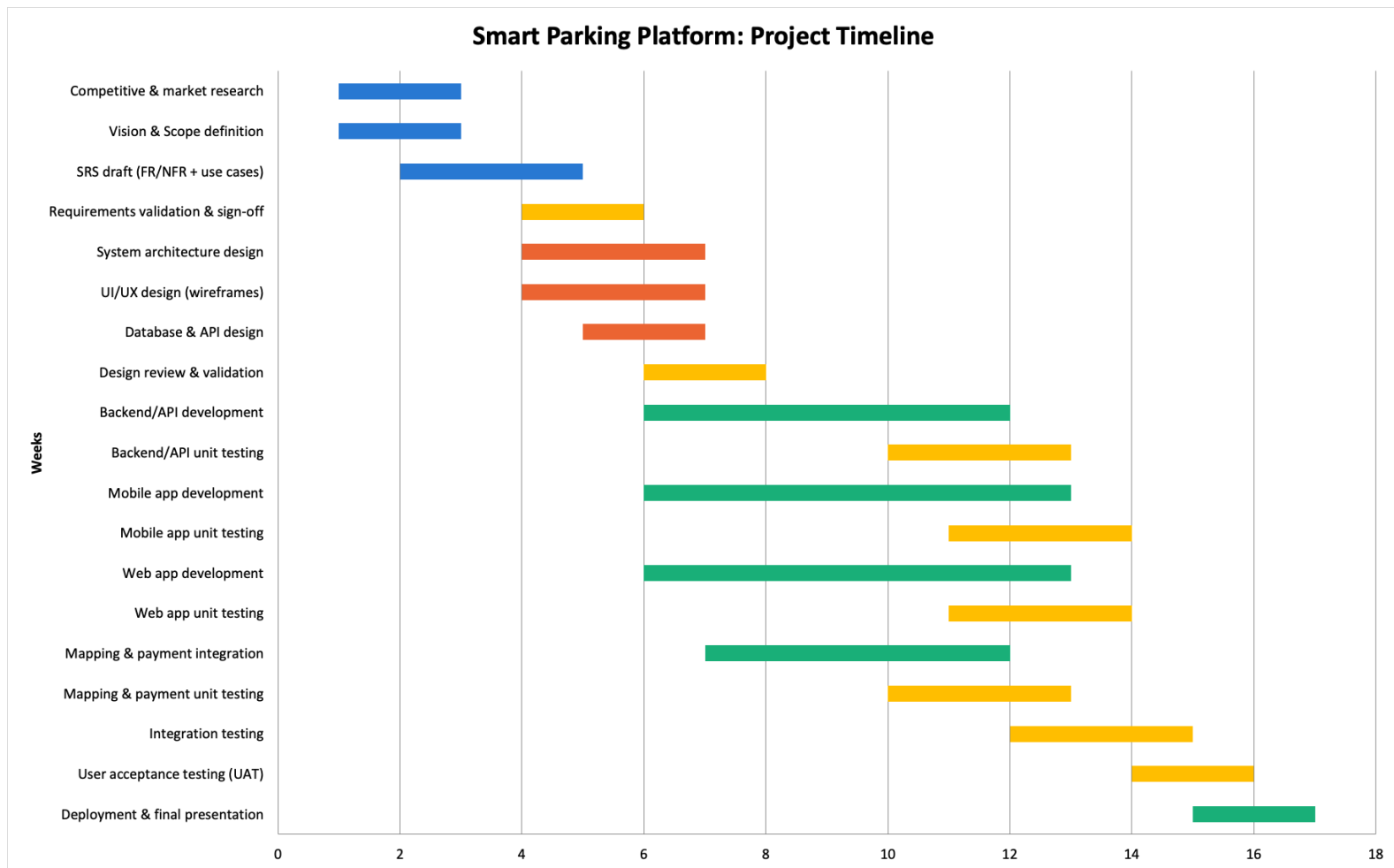
3 Development

- 3.1 Backend/API Development
 - 3.1.1 Build authentication service
 - 3.1.2 Build reservation & availability service
 - 3.1.3 Unit test backend services
- 3.2 Mobile App Development
 - 3.2.1 Build user registration and login flow
 - 3.2.2 Build map & reservation UI
 - 3.2.3 Unit test mobile app modules
- 3.3 Web App Development
 - 3.3.1 Build operator administrative dashboard
 - 3.3.2 Build pricing management interface
 - 3.3.3 Test web app
- 1.3.4 Mapping & Payment Integration
 - 3.4.1 Integrate mapping/navigation API
 - 3.4.2 Integrate payment provider
 - 3.4.3 Unit test integration endpoints

4 Integration, UAT & Deployment

- 4.1 Integration Testing
 - 4.1.1 Test cross-service data flow
 - 4.1.2 Bug triage & fixes
- 4.2 User Acceptance Testing
 - 4.2.1 Conduct UAT sessions with stakeholders
 - 4.2.2 Collect and prioritize feedback
- 4.3 Deployment & Final Presentation
 - 4.3.1 Prepare deployment/launch plan
 - 4.3.2 Finalize project documentation
 - 4.3.3 Deliver final presentation

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AI Usage Disclosure

AI Use Section	Reason
SRS Draft	AI was used in the section of Use Case examples, to help me brainstorm possible uses of the Smart Parking Platform. All use cases were verified, edited, and considered before adding to the document.
WBS Breakdown	Helped draft the section so I could get an understanding of how I should build it.

X. Marcel McNeese Jr.